

Lectures 3 and 4: The Central Limit Theorem

Swapping, the Lindeberg condition, and Stein's method

OUTLINE maximum entropy $\rightarrow$ why a Gaussian at all $\rightarrow$ what a limit in distribution is $\rightarrow$ swap one summand at a time $\rightarrow$ the CLT, and Lindeberg's condition

$\mathbb{E}[Wf(W)] = \mathbb{E}[f'(W)] \rightarrow$ Stein's equation $\rightarrow$ the CLT with a rate $\rightarrow$ the same argument for Poisson

HOW TO READ Sections marked * are not lectured. They are complete, they are examinable, and they carry most of the exercises. Exercises are typed A, B and C; each week's quiz has one of each.

1 Why Gaussians appear, and the CLT by swapping

Q. Why should the Gaussian, of all laws, be the one that appears?

1.1 Notation

Throughout, $X_1, X_2, \dots$ are independent and identically distributed with mean 0 and variance 1, and

$$S_n := \frac{1}{\sqrt{n}} \sum_{i=1}^n X_i.$$

The normalization is chosen so that $\text{Var}(S_n) = 1$ for every n . We write $g \sim \mathcal{N}(0, 1)$, and γ for its law.

From §1.5* of Lectures 1–2. For a random variable X on $\mathbb{R}$ with density p , the *differential entropy* is $h(X) = -\int p \log p$, and the *relative entropy* of X from γ is $D(X\|\gamma) = \int p \log(p/\phi)$, where ϕ is the standard normal density. Both may be $+\infty$, and h is undefined when X has no density.

1.2 What “converging to a Gaussian” means

S_n does not converge to a random variable; what converges is its law, and we test that against functions.

Definition 1.1 (Convergence in distribution). $X_n \Rightarrow X$ means $\mathbb{E}f(X_n) \rightarrow \mathbb{E}f(X)$ for every bounded continuous $f : \mathbb{R} \rightarrow \mathbb{R}$.

Distribution functions need not converge at a jump: if $X_n \equiv 1/n$ and $X \equiv 0$ then $X_n \Rightarrow X$, yet $F_n(0) = 0$ for every n while $F(0) = 1$.

A smooth f can be Taylor expanded and $\mathbf{1}_{(-\infty, z]}$ cannot, so the proofs below test against smooth functions only. That is already enough.

Lemma 1.2. Suppose $\mathbb{E}f(X_n) \rightarrow \mathbb{E}f(X)$ for every $f \in C^3(\mathbb{R})$ with f, f', f'', f''' bounded. Then $F_{X_n}(z) \rightarrow F_X(z)$ at every z where F_X is continuous.

Proof. Fix such a z and $\delta > 0$. Let $\psi : \mathbb{R} \rightarrow [0, 1]$ be smooth with $\psi = 1$ on $(-\infty, 0]$ and $\psi = 0$ on $[1, \infty)$, and put

$$u^+(x) = \psi\left(\frac{x-z}{\delta}\right), \quad u^-(x) = \psi\left(\frac{x-z+\delta}{\delta}\right).$$

Both lie in the test class, and

$$\mathbf{1}_{(-\infty, z-\delta]} \leq u^- \leq \mathbf{1}_{(-\infty, z]} \leq u^+ \leq \mathbf{1}_{(-\infty, z+\delta]}.$$

Taking expectations at X_n and passing to the limit,

$$F_X(z - \delta) \leq \mathbb{E}u^-(X) = \lim_n \mathbb{E}u^-(X_n) \leq \liminf_n F_{X_n}(z),$$

$$\limsup_n F_{X_n}(z) \leq \lim_n \mathbb{E}u^+(X_n) = \mathbb{E}u^+(X) \leq F_X(z + \delta).$$

Let $\delta \downarrow 0$ and use continuity of F_X at z . ■

A rate against smooth f is still not a rate against probabilities; see §2.3.

Exercise 1.3 (TYPE C). Show that the hypothesis of Lemma 1.2 gives the whole of Definition 1.1: $\mathbb{E}f(X_n) \rightarrow \mathbb{E}f(X)$ for every bounded continuous f . *Such an f is neither smooth nor compactly supported; repair each defect in turn, and note that the hypothesis already limits how much mass X_n can put far out.*

1.3 The moral answer

Section 1.5* of Lectures 1–2 supplies two facts. First, at fixed variance the Gaussian maximizes entropy: $h(X) \leq h(g)$ whenever $\text{Var}(X) = 1$. Second, for symmetric i.i.d. X, Y ,

$$h\left(\frac{X+Y}{\sqrt{2}}\right) \geq h(X),$$

because $(x, y) \mapsto \left(\frac{x+y}{\sqrt{2}}, \frac{x-y}{\sqrt{2}}\right)$ is orthogonal, orthogonal maps preserve differential entropy, and entropy is subadditive.

Since $S_{2^k} = \frac{1}{\sqrt{2}}(S'_{2^{k-1}} + S''_{2^{k-1}})$ for two independent copies, $h(S_{2^k})$ increases in k and never exceeds $h(g)$. So it converges, and the maximizer is the only candidate for the limit.

1.4 The entropic CLT

Theorem 1.4 (Barron, 1986). *If $D(S_m \parallel \gamma) < \infty$ for some m , then $D(S_n \parallel \gamma) \rightarrow 0$; equivalently $h(S_n) \rightarrow h(g)$.*

Theorem 1.5 (Artstein–Ball–Barthe–Naor, 2004). *$h(S_n)$ is increasing in n , not merely along powers of two.*

We will not prove either.

Example 1.6 (The entropic CLT says nothing about a coin flip). Let $X_i = \pm 1$ with probability $\frac{1}{2}$ each. Then S_n is supported on a finite set for every n , so it has no density, $h(S_n)$ is undefined, and $D(S_n \parallel \gamma) = \infty$ for every m and n . The hypothesis of Barron’s theorem never holds.

The classical central limit theorem holds for ± 1 regardless, so we need a proof that covers Example 1.6.

1.5 Lindeberg's swapping argument

Theorem 1.7 (Central limit theorem). *Suppose in addition $\mathbb{E}|X_1|^3 < \infty$. Then for every $f \in C^3(\mathbb{R})$ with bounded third derivative,*

$$|\mathbb{E}f(S_n) - \mathbb{E}f(g)| \leq \frac{\|f'''\|_\infty}{6\sqrt{n}} (\mathbb{E}|X_1|^3 + \mathbb{E}|g|^3).$$

Proof. Let $g_1, \dots, g_n$ be i.i.d. standard Gaussians, independent of the X_i , and interpolate:

$$Z_k = \frac{1}{\sqrt{n}}(g_1 + \dots + g_k + X_{k+1} + \dots + X_n), \quad k = 0, \dots, n,$$

so $Z_0 = S_n$ and $Z_n \sim \mathcal{N}(0, 1)$. Telescoping,

$$\mathbb{E}f(S_n) - \mathbb{E}f(g) = \sum_{k=1}^n (\mathbb{E}f(Z_{k-1}) - \mathbb{E}f(Z_k)).$$

Fix k and write $W_k = \frac{1}{\sqrt{n}}(g_1 + \dots + g_{k-1} + X_{k+1} + \dots + X_n)$, which is independent of both X_k and g_k . Then $Z_{k-1} = W_k + X_k/\sqrt{n}$ and $Z_k = W_k + g_k/\sqrt{n}$, and Taylor's theorem gives

$$f(W_k + u) = f(W_k) + uf'(W_k) + \frac{u^2}{2}f''(W_k) + R(u), \quad |R(u)| \leq \frac{\|f'''\|_\infty}{6}|u|^3.$$

Take expectations of both expansions and subtract. The constant terms cancel. The linear terms cancel because W_k is independent of X_k and g_k and both have mean 0. The quadratic terms cancel because both have variance 1. Only the remainders survive:

$$|\mathbb{E}f(Z_{k-1}) - \mathbb{E}f(Z_k)| \leq \frac{\|f'''\|_\infty}{6n^{3/2}} (\mathbb{E}|X_k|^3 + \mathbb{E}|g_k|^3).$$

Summing the n terms gives the claim. ■

The proof used Taylor's theorem and nothing else; in particular it never objected to $X_i = \pm 1$.

With Lemma 1.2, $\mathbb{P}(S_n \leq z) \rightarrow \Phi(z)$ for every z , where Φ is the standard normal distribution function.

The classical argument — expand $\varphi_{X_1}(t/\sqrt{n})$ and take n th powers — is carried out in §1.8*.

1.6 What the proof actually needed

The argument never used that the X_i share a law, nor that they are one-dimensional: the same swap with multivariate Taylor gives the multidimensional central limit theorem. What it did use is that no single summand carries much of the variance.

Definition 1.8 (Lindeberg's condition). Let $\{X_{n,i}\}_{i \leq k_n}$ be a triangular array, independent within rows, with $\mathbb{E}X_{n,i} = 0$ and $s_n^2 = \sum_i \text{Var}(X_{n,i})$. The array satisfies *Lindeberg's condition* if for every $\varepsilon > 0$

$$L_n(\varepsilon) := \frac{1}{s_n^2} \sum_i \mathbb{E} \left[X_{n,i}^2 \mathbf{1}\{|X_{n,i}| > \varepsilon s_n\} \right] \rightarrow 0.$$

Under this condition $s_n^{-1} \sum_i X_{n,i} \Rightarrow \mathcal{N}(0, 1)$, by the same interpolation with the third-moment bound replaced by a truncation.

The condition separates two regimes: many small contributions give a Gaussian, a few rare ones give a Poisson. Section 1.8* computes both sides of the boundary; §2.4 proves the Poisson statement, with a rate.

Remark 1.9 (The limit is universal, the rate is not). Kolmogorov–Smirnov distance from S_{30} to $\mathcal{N}(0, 1)$, by simulation: uniform 0.0014, exponential 0.025, ± 1 0.073. The limit does not depend on the law of X_1 ; the speed does, and lattice summands are slowest.

1.7 * A wide network at initialization

Not lectured. Exercises due at the start of Lecture 4. Fix a nonlinearity $\sigma : \mathbb{R} \rightarrow \mathbb{R}$ and let

$$f_m(x) = \frac{1}{\sqrt{m}} \sum_{j=1}^m a_j \sigma(\langle w_j, x \rangle), \quad x \in \mathbb{R}^d,$$

with $w_1, \dots, w_m$ i.i.d. in $\mathbb{R}^d$ and $a_1, \dots, a_m$ i.i.d. with mean 0 and variance σ_a^2 , independent of the w_j . This is a one-hidden-layer network of width m at initialization, normalized so that $\text{Var } f_m(x)$ does not move as the width grows.

Proposition 1.10. *Fix finitely many inputs $x_1, \dots, x_k$ and suppose $\mathbb{E}[\sigma(\langle w, x_i \rangle)^2] < \infty$ for each i . Then*

$$(f_m(x_1), \dots, f_m(x_k)) \implies \mathcal{N}(0, K), \quad K(x, x') = \sigma_a^2 \mathbb{E}[\sigma(\langle w, x \rangle) \sigma(\langle w, x' \rangle)].$$

Exercise 1.11 (TYPE B). Prove it. Put $V_j = (a_j \sigma(\langle w_j, x_1 \rangle), \dots, a_j \sigma(\langle w_j, x_k \rangle)) \in \mathbb{R}^k$ and identify $(f_m(x_1), \dots, f_m(x_k))$ as $m^{-1/2} \sum_j V_j$. Check $\mathbb{E} V_j = 0$ and $\text{Cov}(V_j) = K$, saying where the independence of a_j from w_j is used in each. Then either quote the multidimensional central limit theorem, or apply Theorem 1.7 to $\langle v, V_j \rangle$ for each fixed $v \in \mathbb{R}^k$ and finish with Corollary 1.7 of Lectures 1–2. Say which of the two routes needs a third moment.

Exercise 1.12 (TYPE A). Show that K is a valid covariance — that $\sum_{i,l} v_i v_l K(x_i, x_l) \geq 0$ for every $v \in \mathbb{R}^k$ — for any σ with $\mathbb{E}[\sigma(\langle w, x_i \rangle)^2] < \infty$. *One line.*

The limit is Gaussian whatever σ and whatever the law of the weights; those enter only through K .

Exercise 1.13 (TYPE C). Let $\sigma = \max(\cdot, 0)$ and $w \sim \mathcal{N}(0, I_d)$. Show that for $x, x' \neq 0$ with angle $\theta = \angle(x, x')$,

$$\mathbb{E}[\sigma(\langle w, x \rangle) \sigma(\langle w, x' \rangle)] = \frac{\|x\| \|x'\|}{2\pi} (\sin \theta + (\pi - \theta) \cos \theta).$$

Check the answer at $\theta = 0$ against $\mathbb{E}[\max(0, g)^2]$, and at $\theta = \pi/2$ against independence. *This is the arccosine kernel.*

Remark 1.14. Proposition 1.10 concerns finitely many inputs. Holding for every finite subset of $\mathbb{R}^d$ at once is what it means to call the limit a *Gaussian process* with kernel K — Lecture 9. Conditioning it on data is Theorem 2.9 of Lectures 1–2 with Σ replaced by the kernel matrix, so an infinitely wide network fitted by exact Bayesian inference is kernel regression with kernel K . At finite width f_m is not Gaussian, and the size of the discrepancy is a question about the *rate* in Proposition 1.10, which Lecture 4 supplies.

1.8 * The proof we did not give

Not lectured. Exercises due at the start of Lecture 4. Characteristic functions turn sums into products, and a product of n nearly-equal factors is an exponential. That is the older proof of Theorem 1.7, and it is worth carrying out once — both to see what it costs, and because the transform does two things at the end of this section that the swap does not.

Lemma 1.15 (Continuity theorem). *Let Y_n and Y be real random variables with $\varphi_{Y_n}(t) \rightarrow \varphi_Y(t)$ for every t . Then $Y_n \Rightarrow Y$.*

Proof. Let f be bounded and continuous with $\hat{f} \in L^1$. By Fourier inversion,

$$\mathbb{E}f(Y_n) = \frac{1}{2\pi} \int_{\mathbb{R}} \hat{f}(\xi) \mathbb{E}[e^{i\xi Y_n}] d\xi = \frac{1}{2\pi} \int_{\mathbb{R}} \hat{f}(\xi) \varphi_{Y_n}(-\xi) d\xi.$$

Since $|\varphi_{Y_n}| \leq 1$ and $\hat{f} \in L^1$, dominated convergence gives $\mathbb{E}f(Y_n) \rightarrow \frac{1}{2\pi} \int \hat{f}(\xi) \varphi_Y(-\xi) d\xi = \mathbb{E}f(Y)$.

The functions used in Exercise 1.3 all lie in this class: a smooth cutoff χ_M is compactly supported, so $\widehat{\chi_M}$ is Schwartz; and $\widehat{g_\delta} = \widehat{g} \widehat{\phi_\delta}$ is a bounded function times a Gaussian. That argument therefore applies verbatim and gives $Y_n \Rightarrow Y$. ■

Exercise 1.16 (TYPE A). For $\mathbb{E}|X|^k < \infty$, differentiate under the integral to get $\varphi_X^{(k)}(0) = (-i)^k \mathbb{E}[X^k]$. Check it against $\varphi(t) = e^{-t^2/2}$ by computing $\varphi''(0)$ and $\varphi^{(4)}(0)$, recovering $\mathbb{E}g^2 = 1$ and $\mathbb{E}g^4 = 3$.

Exercise 1.17 (TYPE B). For $\mathbb{E}X = 0$, $\mathbb{E}X^2 = 1$, show $\varphi_X(t) = 1 - \frac{t^2}{2} + r(t)$ with $r(t)/t^2 \rightarrow 0$. Use $|e^{i\theta} - 1 - i\theta + \theta^2/2| \leq \min(|\theta|^3/6, \theta^2)$ with $\theta = -tX$; the second bound in the minimum is what lets you avoid assuming a third moment.

Exercise 1.18 (TYPE B). Show $(1 + z_n/n)^n \rightarrow e^z$ whenever $z_n \rightarrow z$ in $\mathbb{C}$ (for $|a|, |b| \leq M$, $|a^n - b^n| \leq nM^{n-1}|a - b|$; take $b = e^{z_n/n}$). Combine with the previous exercise to get $\varphi_{S_n}(t) = [\varphi_{X_1}(t/\sqrt{n})]^n \rightarrow e^{-t^2/2}$ and conclude $S_n \Rightarrow \mathcal{N}(0, 1)$. Note where independence was used, and where identical distribution was used.

Exercise 1.19 (TYPE A). Let $B \sim \text{Bernoulli}(p)$; show $\varphi_B(t) = 1 + p(e^{-it} - 1)$. For independent $B_{n,i} \sim \text{Bernoulli}(\lambda/n)$ with $N_n = \sum_i B_{n,i}$, show $\varphi_{N_n}(t) \rightarrow \exp(\lambda(e^{-it} - 1))$ and identify the limit as $\text{Poisson}(\lambda)$ by summing the series. *The same three lines as the previous exercise; only the normalization differs.*

Exercise 1.20 (TYPE B). For the centred array $X_{n,i} = B_{n,i} - \lambda/n$, show $s_n^2 = \lambda(1 - \lambda/n)$ and that $L_n(\varepsilon) \rightarrow 1$ for every $\varepsilon < 1/\sqrt{\lambda}$, so Definition 1.8 fails. Compute $L_n(\frac{1}{2})$ exactly at $\lambda = 3$ for $n = 10, 100, 10^4$; you should get 0, 0.970, 1.000, and the first should be exactly 0 — say why.

Exercise 1.21 (TYPE C). Take $X_{n,i} = \pm 1$ for $i \leq n$ and one extra summand $X_{n,n+1} = \pm\sqrt{n}$, all independent and symmetric. Show $s_n^2 = 2n$, that $L_n(\varepsilon) \geq \frac{1}{2}$ for $\varepsilon < 1/\sqrt{2}$, and that the limiting characteristic function of $s_n^{-1} \sum_i X_{n,i}$ is $e^{-t^2/4} \cos(t/\sqrt{2})$. Exhibit a t at which it vanishes and conclude the limit is not Gaussian. *Why does one zero settle it?*

Exercise 1.22 (TYPE C). The transform also differentiates in the *covariance*. Let f be smooth with $\hat{f} \in L^1$, where $\hat{f}(\xi) = \int f(x) e^{-i\langle x, \xi \rangle} dx$ and $f(x) = (2\pi)^{-d} \int \hat{f}(\xi) e^{i\langle x, \xi \rangle} d\xi$ as in Proposition 1.1 of Lectures 1–2. For a differentiable path $\Sigma(s)$ of covariance matrices and $X_s \sim \mathcal{N}(0, \Sigma(s))$, show

$$\frac{d}{ds} \mathbb{E}[f(X_s)] = \frac{1}{2} \sum_{i,j} \dot{\Sigma}_{ij}(s) \mathbb{E}[\partial_i \partial_j f(X_s)],$$

and integrate along $\Sigma(s) = (1-s)\Sigma^0 + s\Sigma^1$.

This is the exact, continuous form of the swap in Theorem 1.7: interpolate between two Gaussians and the cost is a second derivative. It gives Slepian's comparison inequality in one line when $\Sigma^1 - \Sigma^0$ has non-negative off-diagonal entries and $\partial_i \partial_j f \geq 0$ for $i \neq j$.

Exercise 1.23 (TYPE C). Theorem 2.4 gives rate $n^{-1/2}$. Show that for $X_i = \pm 1$ it cannot be improved. For n even, $\mathbb{P}(S_n = 0) = \binom{n}{n/2} 2^{-n}$, which Stirling puts at $\sim \sqrt{2/\pi n}$; now compare $\mathbb{P}(S_n \leq z)$ with $\Phi(z)$ near $z = 0$. Explain in one sentence why lattice summands were the slowest row of the table in §1.6.

Exercise 1.24 (TYPE C). Show that for every real random variable Y and every $u > 0$,

$$\mathbb{P}(|Y| \geq 2/u) \leq \frac{1}{u} \int_{-u}^u (1 - \varphi_Y(t)) dt.$$

Deduce that if $\varphi_{Y_n} \rightarrow \psi$ pointwise with ψ continuous at 0, then $\sup_n \mathbb{P}(|Y_n| > K) \rightarrow 0$ as $K \rightarrow \infty$. *This is the half of the continuity theorem that does not need the limit law in advance. The other half — that ψ is then the transform of some random variable — is a compactness argument we do not need, since Lemma 1.15 is always applied with the limit already in hand.*

2 Stein's method

Theorem 1.7 describes the Gaussian as a limit. There is another way to pin it down.

Q. Is there an identity satisfied by the standard Gaussian and by no other law?

If so, then showing that a law is approximately Gaussian becomes showing that it approximately satisfies that identity.

2.1 The characterization

Theorem 2.1 (Stein's lemma). *A random variable W is standard Gaussian if and only if*

$$\mathbb{E}[Wf(W)] = \mathbb{E}[f'(W)]$$

for every bounded absolutely continuous f with bounded derivative.

Proof of the forward direction. Let $\phi(x) = (2\pi)^{-1/2} e^{-x^2/2}$, so that $\phi'(x) = -x\phi(x)$. Then

$$\mathbb{E}[f'(g)] = \int f' \phi = - \int f \phi' = \int x f(x) \phi(x) dx = \mathbb{E}[gf(g)].$$

■

The converse is proved in §2.2.

Remark 2.2 (We have used this already). Take $f(x) = e^{-itx}$. Stein's identity becomes $\mathbb{E}[W e^{-itW}] = -it \mathbb{E}[e^{-itW}]$, that is $\varphi'(t) = -t\varphi(t)$, which is the differential equation solved in Lemma 1.10 of Lectures 1–2. The first computation of the course was Stein's lemma with an exponential test function.

Splitting off one summand and Taylor-expanding to reach $\varphi'_{S_n} \approx -t\varphi_{S_n}$ proves the central limit theorem; that route is §1.8*. Exponential test functions give no rate in a probability metric and no tolerance for dependence.

2.2 Stein's equation

Fix a test function h and look for f solving

$$f'(w) - wf(w) = h(w) - \mathbb{E}h(g). \quad (1)$$

Lemma 2.3. Equation (1) is solved by $f_h(w) = e^{w^2/2} \int_{-\infty}^w [h(x) - \mathbb{E}h(g)] e^{-x^2/2} dx$.

Proof. Differentiate: $f'_h(w) = wf_h(w) + [h(w) - \mathbb{E}h(g)]$. ■

Stated without proof. If h is bounded then f_h is bounded and absolutely continuous with $\|f_h\|_\infty \leq \sqrt{2\pi} \|h - \mathbb{E}h(g)\|_\infty$ and $\|f'_h\|_\infty \leq 4\|h - \mathbb{E}h(g)\|_\infty$; if h is moreover Lipschitz then $\|f''_h\|_\infty \leq 2\|h'\|_\infty$. These bounds are the technical core of the method.

Evaluating (1) at W and taking expectations turns a distance between two laws into a single expectation:

$$\mathbb{E}h(W) - \mathbb{E}h(g) = \mathbb{E}[f'_h(W) - Wf_h(W)]. \quad (2)$$

This also settles the converse in Theorem 2.1. If W satisfies the identity, then for every bounded Lipschitz h the right side of (2) vanishes, since f_h is an admissible test function; so $\mathbb{E}h(W) = \mathbb{E}h(g)$ for all such h , and hence $W \stackrel{d}{=} g$.

2.3 The central limit theorem, with a rate

Theorem 2.4. Let $X_1, \dots, X_n$ be i.i.d. with mean 0, variance 1 and $\mathbb{E}|X_1|^3 < \infty$. Then for every bounded Lipschitz h ,

$$|\mathbb{E}h(S_n) - \mathbb{E}h(g)| \leq \frac{C \|h'\|_\infty \mathbb{E}|X_1|^3}{\sqrt{n}}$$

for an absolute constant C .

Sketch. Write $W = S_n$ and $W^{(i)} = W - X_i/\sqrt{n}$, which is independent of X_i . Abbreviate $f = f_h$. Then

$$\mathbb{E}[Wf(W)] = \frac{1}{\sqrt{n}} \sum_i \mathbb{E}[X_i f(W)],$$

and expanding $f(W)$ about $W^{(i)}$ gives, for each i ,

$$\mathbb{E}[X_i f(W)] = \underbrace{\mathbb{E}[X_i] \mathbb{E}[f(W^{(i)})]}_{=0} + \frac{1}{\sqrt{n}} \underbrace{\mathbb{E}[X_i^2]}_{=1} \mathbb{E}[f'(W^{(i)})] + \text{remainder},$$

with the remainder bounded by $\|f''\|_\infty \mathbb{E}|X_i|^3/n$. Summing over i turns the middle terms into $\mathbb{E}f'(W)$ up to an error of the same order, so

$$|\mathbb{E}[f'(W) - Wf(W)]| \leq \frac{C' \|f''\|_\infty \mathbb{E}|X_1|^3}{\sqrt{n}},$$

and (2) with the bound $\|f''_h\|_\infty \leq 2\|h'\|_\infty$ finishes the proof. ■

Remark 2.5. Theorem 1.7 produced the rate $n^{-1/2}$ as a by-product of its error term; here the rate is what the method computes. Taking h to approximate $\mathbf{1}_{(-\infty, z]}$, and controlling the approximation, upgrades Theorem 2.4 to the Berry–Esseen theorem: $\sup_z |\mathbb{P}(S_n \leq z) - \mathbb{P}(g \leq z)| \leq C \mathbb{E}|X_1|^3/\sqrt{n}$. The extra work is in the indicator, not in the method.

2.4 The same argument for Poisson

Nothing above was specific to the Gaussian; only the characterizing identity was used.

Theorem 2.6 (Chen's lemma). *A random variable N taking values in $\{0, 1, 2, \dots\}$ is $\text{Poisson}(\lambda)$ if and only if*

$$\mathbb{E}[\lambda f(N+1) - Nf(N)] = 0$$

for every bounded f .

Proof of the forward direction. $\mathbb{E}[Nf(N)] = \sum_{k \geq 1} k e^{-\lambda} \frac{\lambda^k}{k!} f(k) = \lambda \sum_{j \geq 0} e^{-\lambda} \frac{\lambda^j}{j!} f(j+1) = \lambda \mathbb{E}[f(N+1)]$. ■

Running §2.2 and §2.3 with this identity in place of Stein's, and total variation in place of a smooth metric, gives:

Theorem 2.7 (Le Cam). *Let $B_1, \dots, B_n$ be independent with $B_i \sim \text{Bernoulli}(p_i)$, and let $\lambda = \sum_i p_i$. Then*

$$d_{\text{TV}}\left(\sum_i B_i, \text{Poisson}(\lambda)\right) \leq \sum_i p_i^2.$$

Corollary 2.8 (Poisson limit theorem). *Let $B_{n,i} \sim \text{Bernoulli}(p_{n,i})$, $i \leq k_n$, be independent, with $\max_i p_{n,i} \rightarrow 0$ and $\sum_i p_{n,i} \rightarrow \lambda$. Then $\sum_i B_{n,i} \Rightarrow \text{Poisson}(\lambda)$.*

Proof. $\sum_i p_{n,i}^2 \leq (\max_i p_{n,i}) \sum_i p_{n,i} \rightarrow 0$, and $\text{Poisson}(\lambda_n) \Rightarrow \text{Poisson}(\lambda)$ when $\lambda_n \rightarrow \lambda$. ■

This is the Poisson side of the Lindeberg boundary. At $p_i = \lambda/n$ the rate is λ^2/n ; with $\lambda = 3$ and $n = 100$ the bound is 0.09, against an actual discrepancy of about 0.002 in $\mathbb{P}(\sum_i B_i = 0)$.

Lecture 3 met the Gaussian/Poisson boundary as a phenomenon. Here a single technique sits on both sides of it, and only the characterizing identity changes.

2.5 * Where independence was spent

Not lectured. Exercises due at the start of Lecture 5. The proof of Theorem 2.4 used independence in exactly one place, to evaluate $\mathbb{E}[X_i f(W^{(i)})] = \mathbb{E}[X_i] \mathbb{E}[f(W^{(i)})]$. Any other route to that expectation will do, and the cheapest is a coupling.

Definition 2.9 (Exchangeable pair). A pair (W, W') is exchangeable if $(W, W') \stackrel{d}{=} (W', W)$. It is a *Stein pair* if in addition $\mathbb{E}[W' - W \mid W] = -aW$ for some $a > 0$.

Exercise 2.10 (TYPE A). Let (W, W') be exchangeable and F antisymmetric, $F(x, y) = -F(y, x)$, with $\mathbb{E}|F(W, W')| < \infty$. Show $\mathbb{E}F(W, W') = 0$. *One line, and it is the whole mechanism; everything else is choosing F .*

Exercise 2.11 (TYPE B). Apply that to $F(x, y) = (y - x)(f(x) + f(y))$, expand $f(W')$ about W to first order, and use Definition 2.9 to obtain

$$\mathbb{E}[f'(W) - Wf(W)] = \mathbb{E}\left[f'(W)\left(1 - \frac{1}{2a}\mathbb{E}[(W' - W)^2 \mid W]\right)\right] + \text{err}, \quad |\text{err}| \leq \frac{\|f''\|_\infty}{4a} \mathbb{E}|W' - W|^3.$$

A Gaussian approximation now asks two things: that $\frac{1}{2a}\mathbb{E}[(W' - W)^2 \mid W]$ be close to 1, and that $W' - W$ be small.

Exercise 2.12 (TYPE B). *The canonical pair.* With $W = S_n$, let I be uniform on $\{1, \dots, n\}$ and X'_I an independent copy of X_I ; put $W' = W - (X_I - X'_I)/\sqrt{n}$. Show (W, W') is exchangeable with $\mathbb{E}[W' - W \mid X_1, \dots, X_n] = -W/n$, so $a = 1/n$; then that $\mathbb{E}[(W' - W)^2 \mid X_1, \dots, X_n] = n^{-2} \sum_i (X_i^2 + 1)$, whose distance from 1 after dividing by $2a$ is at most $\frac{1}{2} \sqrt{\text{Var}(X_1^2)}/n$. Conclude Theorem 2.4 again. *Independence was used twice; neither use was the product formula.*

Now a sum to which no version of independence applies. Let π be a uniformly random permutation of $\{1, \dots, n\}$, let $I_i = \mathbf{1}\{\pi(i) = i\}$, and let $W = \sum_i I_i$ count fixed points.

Exercise 2.13 (TYPE A). Show $\mathbb{E}W = \text{Var}W = 1$ for $n \geq 2$, and that $\mathbb{P}(I_1 = I_2 = 1) \neq \mathbb{P}(I_1 = 1)\mathbb{P}(I_2 = 1)$. A student applies Theorem 2.7 with $p_i = 1/n$ and $\lambda = 1$ to conclude $d_{\text{TV}}(W, \text{Poisson}(1)) \leq 1/n$. Name the hypothesis that fails, and say why the conclusion happening to be true does not rescue the argument.

Imported: Chen–Stein, coupling form. Let $W = \sum_{i \leq n} I_i$ with $I_i \sim \text{Bernoulli}(p_i)$, *not necessarily independent*, and $\lambda = \sum_i p_i$. If for each i one can build, on the same space as W , a variable V_i distributed as $W - 1$ conditioned on $I_i = 1$, then $d_{\text{TV}}(W, \text{Poisson}(\lambda)) \leq \min(1, \lambda^{-1}) \sum_i p_i \mathbb{E}|W - V_i|$.

Exercise 2.14 (TYPE C). Build the coupling. Fix i ; if $\pi(i) = i$ put $\sigma = \pi$, otherwise $\sigma = \pi \circ (i \ \pi^{-1}(i))$. Show $\sigma(i) = i$ always, that σ is uniform among permutations fixing i , and that $W(\sigma) - W(\pi) \in \{1, 2\}$ when $\pi(i) \neq i$ — the value 2 exactly when π has a 2-cycle at i . Deduce $\mathbb{E}|W - V_i| \leq 2/n$, hence $d_{\text{TV}}(W, \text{Poisson}(1)) \leq 2/n$.

Inclusion–exclusion gives $\mathbb{P}(W = 0) = \sum_{k \leq n} (-1)^k / k!$, whose distance from e^{-1} is 7.1×10^{-3} , 1.8×10^{-4} and 2.3×10^{-8} at $n = 4, 6, 10$. *By how many orders of magnitude is the method wrong at $n = 10$, and what does it have that inclusion–exclusion does not?*

§1.7* carries out the wide-network limit and computes the arccosine kernel; §1.8* gives the characteristic-function proof of the central limit theorem, the same computation for Poisson, and Lindeberg’s condition on both sides of the boundary; §2.5* develops the exchangeable-pair bound and a case with no independence at all.